Rational function approximation

Rational function of degree $N = n + m$ is written as

$$r(x) = \frac{p(x)}{q(x)} = \frac{p_0 + p_1 x + \cdots + p_n x^n}{q_0 + q_1 x + \cdots + q_m x^m}$$

Now we try to approximate a function f on an interval containing 0 using $r(x)$.

WLOG, we set $q_0 = 1$, and will need to determine the $N + 1$ unknowns $p_0, \dots, p_n, q_1, \dots, q_m$.

Padé approximation

The idea of **Padé approximation** is to find $r(x)$ such that

$$f^{(k)}(0) = r^{(k)}(0), \quad k = 0, 1, \dots, N$$

This is an extension of Taylor series but in the rational form.

Denote the Maclaurin series expansion $f(x) = \sum_{i=0}^{\infty} a_i x^i$. Then

$$f(x) - r(x) = \frac{\sum_{i=0}^{\infty} a_i x^i \sum_{i=0}^{m} q_i x^i - \sum_{i=0}^{n} p_i x^i}{q(x)}$$

If we want $f^{(k)}(0) - r^{(k)}(0) = 0$ for $k = 0, \dots, N$, we need the numerator to have 0 as a root of multiplicity $N + 1$.

Padé approximation

This turns out to be equivalent to

$$\sum_{i=0}^{k} a_i q_{k-i} = p_k, \quad k = 0, 1, \dots, N$$

for convenience we used convention $p_{n+1} = \cdots = p_N = 0$ and $q_{m+1} = \cdots = q_N = 0$.

From these $N+1$ equations, we can determine the $N+1$ unknowns:

$$p_0, p_1, \dots, p_n, q_1, \dots, q_m$$

Padé approximation

Example

Find the Padé approximation to e^{-x} of degree 5 with $n = 3$ and $m = 2$.

Solution. We first write the Maclaurin series of e^{-x} as

$$e^{-x} = 1 - x + \frac{1}{2}x^2 - \frac{1}{6}x^3 + \frac{1}{24}x^4 + \cdots = \sum_{i=0}^{\infty} \frac{(-1)^i}{i!}x^i$$

Then for $r(x) = \frac{p_0+p_1x+p_2x^2+p_3x^3}{1+q_1x+q_2x^2}$, we need

$$\left(1 - x + \frac{1}{2}x^2 - \frac{1}{6}x^3 + \cdots\right)(1+q_1x+q_2x^2)-p_0+p_1x+p_2x^2+p_3x^3$$

to have 0 coefficients for terms $1, x, \dots, x^5$.

Padé approximation

Solution. (cont.) By solving this, we get p_0, p_1, p_2, q_1, q_2 and hence

$$r(x) = \frac{1 - \frac{3}{5}x + \frac{3}{20}x^2 - \frac{1}{60}x^3}{1 + \frac{2}{5}x + \frac{1}{20}x^2}$$

x	e^{-x}	$P_5(x)$	$\lvert e^{-x} - P_5(x)\rvert$	$r(x)$	$\lvert e^{-x} - r(x)\rvert$
0.2	0.81873075	0.81873067	8.64×10^{-8}	0.81873075	7.55×10^{-9}
0.4	0.67032005	0.67031467	5.38×10^{-6}	0.67031963	4.11×10^{-7}
0.6	0.54881164	0.54875200	5.96×10^{-5}	0.54880763	4.00×10^{-6}
0.8	0.44932896	0.44900267	3.26×10^{-4}	0.44930966	1.93×10^{-5}
1.0	0.36787944	0.36666667	1.21×10^{-3}	0.36781609	6.33×10^{-5}

where $P_5(x)$ is Maclaurin polynomial of degree 5 for comparison.

Chebyshev rational function approximation

To obtain more uniformly accurate approximation, we can use Chebyshev polynomials $T_k(x)$ in Padé approximation framework.

For $N = n + m$, we use

$$r(x) = \frac{\sum_{k=0}^{n} p_k T_k(x)}{\sum_{k=0}^{m} q_k T_k(x)}$$

where $q_0 = 1$. Also write $f(x)$ using Chebyshev polynomials as

$$f(x) = \sum_{k=0}^{\infty} a_k T_k(x)$$

Chebyshev rational function approximation

Now we have

$$f(x) - r(x) = \frac{\sum_{k=0}^{\infty} a_k T_k(x) \sum_{k=0}^{m} q_k T_k(x) - \sum_{k=0}^{n} p_k T_k(x)}{\sum_{k=0}^{m} q_k T_k(x)}$$

We again seek for $p_0, \dots, p_n, q_1, \dots, q_m$ such that coefficients of $1, x, \dots, x^N$ are 0.

To that end, the computations can be simplified due to

$$T_i(x) T_j(x) = \frac{1}{2} \Big(T_{i+j}(x) + T_{|i-j|}(x) \Big)$$

Also note that we also need to compute Chebyshev series coefficients a_k first.

Chebyshev rational function approximation

Example

Approximate e^{-x} using the Chebyshev rational approximation of degree $n = 3$ and $m = 2$. The result is $r_T(x)$.

x	e^{-x}	$r(x)$	$\lvert e^{-x} - r(x)\rvert$	$r_T(x)$	$\lvert e^{-x} - r_T(x)\rvert$
0.2	0.81873075	0.81873075	7.55×10^{-9}	0.81872510	5.66×10^{-6}
0.4	0.67032005	0.67031963	4.11×10^{-7}	0.67031310	6.95×10^{-6}
0.6	0.54881164	0.54880763	4.00×10^{-6}	0.54881292	1.28×10^{-6}
0.8	0.44932896	0.44930966	1.93×10^{-5}	0.44933809	9.13×10^{-6}
1.0	0.36787944	0.36781609	6.33×10^{-5}	0.36787155	7.89×10^{-6}

where $r(x)$ is the standard Padé approximation shown earlier.

Trigonometric polynomial approximation

Recall the Fourier series uses a set of $2n$ orthogonal functions with respect to weight $w \equiv 1$ on $[-\pi, \pi]$:

$$\begin{aligned}
\phi_0(x) &= \frac{1}{2} \\
\phi_k(x) &= \cos kx, \quad k = 1, 2, \dots, n \\
\phi_{n+k}(x) &= \sin kx, \quad k = 1, 2, \dots, n-1
\end{aligned}$$

We denote the set of linear combinations of $\phi_0, \phi_1, \dots, \phi_{2n-1}$ by $\mathcal{T}_n$, called the set of trigonometric polynomials of degree $\leq n$.

Trigonometric polynomial approximation

For a function $f \in C[-\pi, \pi]$, we want to find $S_n \in \mathcal{T}_n$ of form

$$S_n(x) = \frac{a_0}{2} + a_n \cos nx + \sum_{k=1}^{n-1} (a_k \cos kx + b_k \sin kx)$$

to minimize the least squares error

$$E(a_0, \dots, a_n, b_1, \dots, b_{n-1}) = \int_{-\pi}^{\pi} |f(x) - S_n(x)|^2 \, \mathrm{d}x$$

Due to orthogonality of Fourier series $\phi_0, \dots, \phi_{2n-1}$, we get

$$a_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos kx \, \mathrm{d}x, \quad b_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin kx \, \mathrm{d}x$$

Trigonometric polynomial approximation

Example

Approximate $f(x) = |x|$ for $x \in [-\pi, \pi]$ using trigonometric polynomial from $\mathcal{T}_n$.

Solution. It is easy to check that $a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} |x| \,\mathrm{d}x = \pi$ and

$$a_k = \frac{1}{\pi}\int_{-\pi}^{\pi} |x| \cos kx \,\mathrm{d}x = \frac{2}{\pi k^2}((-1)^k - 1), \quad k = 1, 2, \dots, n$$

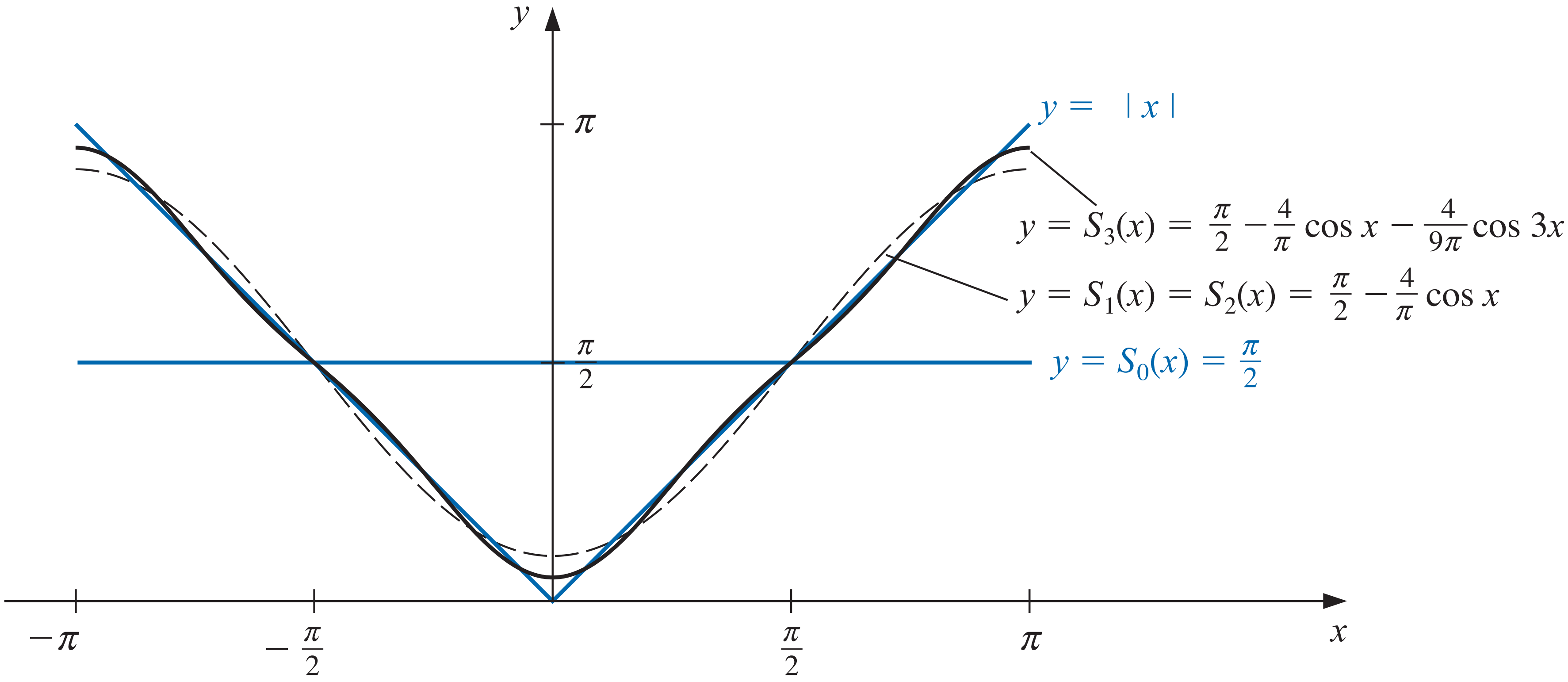
$$b_k = \frac{1}{\pi}\int_{-\pi}^{\pi} |x| \sin kx \,\mathrm{d}x = 0, \quad k = 1, 2, \dots, n-1$$

Therefore

$$S_n(x) = \frac{\pi}{2} + \frac{2}{\pi}\sum_{k=1}^{n} \frac{(-1)^k - 1}{k^2} \cos kx$$

Trigonometric polynomial approximation

$S_n(x)$ for the first few n are shown below:



Discrete trigonometric approximation

If we have $2m$ paired data points $\{(x_j, y_j)\}_{j=0}^{2m-1}$ where x_j are equally spaced on $[-\pi, \pi]$, i.e.,

$$x_j = -\pi + \left(\frac{j}{m}\right)\pi, \quad j = 0, 1, \dots, 2m-1$$

Then we can also seek for $S_n \in \mathcal{T}_n$ such that the discrete least square error below is minimized:

$$E(a_0, \dots, a_n, b_1, \dots, b_{n-1}) = \sum_{j=0}^{2m-1} (y_j - S_n(x_j))^2$$

Discrete trigonometric approximation

Theorem

Define

$$a_k = \frac{1}{m}\sum_{j=0}^{2m-1} y_j \cos kx_j, \quad b_k = \frac{1}{m}\sum_{j=0}^{2m-1} y_j \sin kx_j$$

Then the trigonometric $S_n \in \mathcal{T}_n$ defined by

$$S_n(x) = \frac{a_0}{2} + a_n \cos nx + \sum_{k=1}^{n-1}(a_k \cos kx + b_k \sin kx)$$

minimizes the discrete least squares error

$$E(a_0, \ldots, a_n, b_1, \ldots, b_{n-1}) = \sum_{j=0}^{2m-1} (y_j - S_n(x_j))^2$$

Fast Fourier transforms

The **fast Fourier transform (FFT)** employs the Euler formula $e^{z\mathrm{i}} = \cos z + \mathrm{i} \sin z$ for all $z \in \mathbb{R}$ and $\mathrm{i} = \sqrt{-1}$, and compute the discrete Fourier transform of data to get

$$\frac{1}{m} \sum_{k=0}^{2m-1} c_k e^{kx\mathrm{i}}, \text{ where } c_k = \sum_{j=0}^{2m-1} y_j e^{k\pi \mathrm{i}/m} \quad k = 0, \dots, 2m-1$$

Then one can recover $a_k, b_k \in \mathbb{R}$ from

$$a_k + \mathrm{i} b_k = \frac{(-1)^k}{m} c_k \in \mathbb{C}$$

Fast Fourier transforms

The discrete trigonometric approximation for $2m$ data points requires a total of $(2m)^2$ multiplications, not scalable for large m.

The cost of FFT is only

$$3m + m\log_2 m = O(m\log_2 m)$$

For example, if $m = 1024$, then $(2m)^2 \approx 4.2 \times 10^6$ and $3m + m\log_2 m \approx 1.3 \times 10^4$. The larger m is, the more benefit FFT gains.